CareBridge: Human-Centered Emergency Care and Classical AI for Emergency-Care Coordination

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Abstract

Emergency-care operations require coordination of patient intake, observations, clinical assessment, staff, rooms, equipment, and changing care needs. This paper presents CareBridge, a student-developed software prototype for emergency-care coordination using explicit workflow rules and classical artificial-intelligence concepts. The design separates recorded observations, system-generated indicators, doctor-entered assessment, operational priority, and resource allocation. New observations are preserved as historical events, while relevant changes refresh operational priorities and alerts according to documented policy. Resource planning may use priority queues, matching, constraint satisfaction, or A* search when explicitly formulated. Clinical discharge remains a doctor-led decision followed by administrative verification. The supplied development report states that 106 software tests passed and build/browser checks succeeded; these are reported implementation checks, not clinical validation. The prototype is not presented as clinically validated or production-ready.

Keywords— Emergency-care coordination, classical AI, resource allocation, rule-based system, priority queue, constraint satisfaction, A* search, auditability.

1. Introduction

Emergency departments coordinate interdependent activities under time pressure. Observations change, clinicians reassess patients, resources become unavailable, and operational priorities must reflect current information. When observations, clinician decisions, and operational state are conflated, users may not understand why a patient has a particular status or why a resource was assigned. CareBridge is a human-centred coordination prototype that organises workflow and resource information while preserving clinician authority. Classical AI refers here to explicit rules, search, prioritisation, and constraint-based reasoning—not an assumed black-box predictive model.

1.1 Objectives and scope

Objectives are to provide role-based workspaces; preserve time-stamped observation history; separate system indicators from doctor-entered assessment; refresh operational priority and alerts when relevant information changes; represent resource constraints explicitly; and maintain an auditable, doctor-led discharge workflow. The scope is limited to software coordination and verification. No real-hospital deployment, clinical trial, patient-outcome benefit, or clinical validation is claimed.

2. Background and Related Work

Early-warning frameworks organise physiological observations into structured indicators and escalation processes. The Royal College of Physicians' NEWS2 report describes a standardised approach to recognising and responding to acute illness [1]. This supports consistent observation and escalation design, but does not establish that a student prototype implements a clinically equivalent score. Rajkomar, Dean, and Kohane discuss machine learning in medicine, including data quality, evaluation, workflow integration, and clinical context [2]. Future predictive models would require suitable data, validation, governance, and monitoring. A* search combines path cost and a heuristic estimate to search for a least-cost path under appropriate conditions [3]. It should be claimed only when states, actions, costs, a goal condition, and a justified heuristic are defined. Resource assignment may instead suit matching, constraint satisfaction, or optimisation.

3. Proposed Method

3.1 System architecture

The logical architecture contains a role-based web interface, application/API layer, workflow and policy services, planning/allocation component, persistent data layer, and audit log. Authentication and authorisation are enforced server-side; workflow transitions are validated centrally.

Layer	Responsibility
Role UI	Receptionist, doctor, administrator workspaces
API/workflow	Authentication, validation, state transitions
Policy services	Indicators, priority refresh, alerts, discharge checks
Planning	Resource constraints; queue, matching, CSP or defined search
Persistence/audit	Events, assessments, resource state, actor/timestamp history

Table 1. Logical architecture and responsibilities.

3.2 State and reassessment

The data model distinguishes observations, system-generated indicators, doctor-entered assessment, operational priority, and resource allocation. These values are related but not interchangeable. Disagreement between system output and doctor assessment should remain visible and follow a defined review policy. Each new observation should be stored as a separate time-stamped event; earlier measurements should not be overwritten. Relevant changes should refresh dependent indicators, priority, and alerts, with the reason and time of each consequential change shown to users.

3.3 Resource allocation and discharge

Allocation constraints may include staff availability, room capability, equipment, patient requirements, and urgency. Priority queues, matching, CSP, or optimisation should be selected based on the actual formulation. A doctor records a clinical discharge request; the receptionist performs administrative verification and completes discharge. Resources are released after final completion. A lower system indicator or stable status does not independently authorise discharge.

4. Implementation and Verification

Project notes describe a database-backed web application with separate receptionist, doctor, and administrator workspaces. A reproducible submission should specify framework versions, schema, repository revision, environment requirements, seed-data procedure, and test commands.

4.1 Verification strategy

Verification should cover policy rules and state transitions, authentication and role permissions, persistence, resource allocation, and end-to-end journeys. Edge cases include missing observations, disagreement between system and doctor assessment, unavailable resources, duplicate requests, failed persistence, and unauthorised transitions.

4.2 Reported verification

The latest development report supplied for this project states that 106 tests passed with zero failures and that build and browser checks succeeded. These results are reported by the development workflow and have not been independently reproduced for this paper. Retain the relevant test logs and build output.

Interpretation: passing software tests supports only the behaviour covered by those tests. It does not establish clinical accuracy, patient safety, regulatory compliance, or improved outcomes.

5. Results and Discussion

The supplied report indicates successful software verification and browser checks for the tested revision. Evidence strength depends on test coverage, retained logs, and whether the submitted repository matches the tested revision. A key acceptance criterion is state consistency: when an observation or doctor assessment changes, dependent operational fields should refresh according to policy and the change should be traceable. A lower risk indicator paired with an unexplained status or priority should be treated as a consistency defect until resolved.

5.1 Limitations

No clinical trial, clinical validation, or patient-outcome benefit is reported. The test count comes from supplied development notes rather than independent reproduction. Clinical thresholds and rules would require qualified clinical review before real-world use. Allocation quality depends on accurate resource data and explicit constraints.

5.2 Future evaluation

Future evaluation may measure workflow completion time, state-transition consistency, allocation feasibility, resource utilisation, alert latency, audit completeness, and user comprehension. Metrics and scenarios should be defined before testing. No numerical results are claimed because an evaluation dataset and protocol were not provided.

6. Conclusion

CareBridge presents a human-centred coordination design using role-based workflows, event history, explicit state separation, rule-based indicators, and classical planning concepts. Its core principle is to keep observations, system indicators, clinician assessment, operational priority, and resource allocation distinct while ensuring that updates propagate consistently and remain auditable. The reported 106 passing tests and build/browser checks support software verification only; they do not establish clinical validity, safety, or hospital readiness. Future work should focus on state-consistency testing, documented allocation formulations, stakeholder-reviewed workflows, usability evaluation, and security hardening.

References

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